LOK JAGRUTI UNIVERSITY (LJU)

INSTITUTE OF ENGINEERING & TECHNOLOGY

Department of CE, IT, CSD, AIML, AIDS, CSE, CST, CSIT, CEA (701, 702, 703, 704, 705, 712, 713, 714, 715)

Bachelor of Technology (B.E.) – Semester - IV

Course Code:	**117013492, 117023492, 117033492, 117043492, 117053492, 117123492, 117133492, 117143492, 117153492,**
Course Name:	**Full Stack Development with Javascript-2**
Category of Course:	Engineering Science Course (ESC)
Prerequisite Course:	Full Stack Development with Javascript-1(117012191)

Teaching Scheme				
Lecture (L)	**Tutorial (T)**	**Practical (P)**	**Credit**	**Total Hours**
3	**0**	**6**	**6**	**30**

Syllabus

Unit No.	Topic	Prerequisite Topic	Successive Topic	Teaching Hours
01	**JSON**			**1 (5%)**
	1.1 JSON Fundamentals - JSON Introduction, JSON Syntax and Structure; JSON Data Types: Number, String, Boolean, Null, Object, Array **1.2 JSON Methods-** JSON.parse(), JSON.stringify()	---	2.1 Introduction to Node.js	
	1.3 Working with JSON Objects and Arrays; Nested JSON, Accessing & Modifying Data	---	---	
02	**Node JS- Introduction and Core modules**			**3 (10%)**
	2.1 Introduction to Node.js - Introduction to Node.js, Node.js Architecture, Event Loop & Event-Driven Model, Installation & Setup, Asynchronous Programming (Callbacks & Non-blocking I/O), NPM & Package Management, Environment Variables Setup	1.1 JSON Fundamentals	10.3 Advanced Concepts - Full MERN Stack Connectivity	
	2.2 Core Modules - File System Module (Synchronous & Asynchronous Methods), Path Module, Events Module & EventEmitter	---	---	
03	**Node JS Modules & Server Creation**			**3 (10%)**
	3.1 URL Module**,** Introduction to HTTP Module, Creating HTTP Server, Handling Requests & Responses, Basic Routing **3.2** Serving HTML Files, Sending JSON Response	(117013392 – Unit 1) 1.1 Basics of WWW, HTTP Protocol	4.1 Introduction to Express.js	
04	**Express JS Fundamentals**			**2 (5%)**
	4.1 Introduction to Express.js - Introduction to Express.js, Features & Advantages, Installation & Project Setup, Understanding app.use() and app.listen()	3.1 URL Module**,** Introduction to HTTP Module	10.3 Advanced Concepts - Full MERN Stack Connectivity	
	4.2 Routing & Middleware - Express Routing using HTTP methods, Route Parameters, Query Parameters, Middleware	(117013392 – Unit 8**)** 8.1 Basics of Javascript and EcmaScript(ES6), 8.2 Loops & Conditions	---	
05	**Express State Management and API**			**3 (10%)**
	5.1 Static Files & State Management - Serving Static Files (express.static), Handling Forms & Form Data, Cookies, Session Management	(117013392 – Unit 3**)** 3.1 Form & Input Tags	---	
	5.2 RESTful APIs - REST Architecture & Principles, Designing RESTful APIs using Express, Basics of CORS	---	8.2 API Integration 10.3 Advanced Concepts - Full MERN Stack Connectivity	
06	**Express – Advanced Concept**			**3 (10%)**
	6.1 File Handling & Data Processing - File Upload using Multer, Handling Multipart/Form Data	(117013392 – Unit 3**)** 3.1 Form & Input Tags	---	
	6.2 Advanced Express Concepts -Template Engine (EJS), Emails using Nodemailer	---	---	
07	**React Fundamentals & Core Concepts**			**4 (10%)**
	7.1 Introduction & Setup - Introduction to React, Advantages, SPA Concept, Installation & Setup using Vite, Folder Structure, Understanding package.json, React Rendering	---	---	

	7.2 Core Concepts - Virtual DOM Concept, JSX Syntax & Rules, JSX Attributes & Styling, Functional Components, Props, Event Handling	(117013392 – Unit 8,9**)** 8.1 Basics of Javascript and EcmaScript(ES6), 8.2 Loops & Conditions 9.2 Event Handling	---	
	7.3 React Routing - Introduction to React Router, BrowserRouter, Routes & Route	(117013392 – Unit 10**)** 10.2 Asynchronous JavaScript:	---	
08	**React Hooks & API Integration**			**4 (15%)**
	8.1 React Hooks - useState, useEffect, useReducer, useContext	(117013392 – Unit 10**)** 10.3 Explain the fetch API for making HTTP requests to external APIs	---	
	8.2 Forms & API Integration - Handling Forms, Axios for API Integration	(117013392 – Unit 3,10**)** 3.1 Form & Input Tags 10.3 Explain the fetch API 5.2 RESTful APIs	10.3 Advanced Concepts – Full MERN Stack Connectivity	
09	**MongoDB- Queries and operators**			**3 (10%)**
	9.1 Introduction to MongoDB - NoSQL Databases, Documents & Collections, Difference between RDBMS and NoSQL, MongoDB vs MySQL, Installation & Setup	---	---	
	9.2 Query & Operators - Comparison Operators , Logical Operators ,Update Operators ,Projection, Sorting, Limiting Records	---	---	
	9.3 Aggregation Framework - Introduction to Aggregation, $match, $group, $sort, $project			
10	**Mongoose & MERN Integration**			**4 (15%)**
	10.1 Introduction to Mongoose - What is Mongoose, Installing & Connecting with Node/Express, Creating Schema & Model, Data Types, Schema Validation	---	---	
	10.2 CRUD Operations using Mongoose - Create, Read, Update, Delete	---	---	
	10.3 Advanced Concepts – Indexing, Regular Expressions (RegEx), **Full MERN Stack Connectivity**	(117013392 – Unit 10**)** 10.1 Form Validation, 2.1 Introduction to Node.js 4.1 Introduction to Express.js 5.2 Basics of CORS 8.2 Forms & API Integration	---	

Sr No.	Practical Title	Link to Theory Syllabus
1	Write a function 'FirstAndLast' that takes in an array, and returns an object with: 1) the first element of the array as the object's key, and 2) the last element of the array as that key's value. (Example input: ['ABC', 'DEF', 'Employee', 'Manager'] output: ABC : 'Manager')	**Unit-1**
2	Print following statement in console from given JSON objects. const a = {"Name" : "Ramesh", "Subects" : ["Maths", "Scence" , "chemistry"], "Grade" : {"Type" : "marks","Total" : [88,90,99,87]}, "Range" : {"opt" : "100", "type" : ["secure","subject","class"]}, "achive" : [{"Rank" :"rank", "place":[1,2,3]}, {"Ordinalindicator":"st"},"12"], "joints" : ['outof','got','and'] } Output: Ramesh got 99 outof 100 marks and secure 1st rank.	**Unit-1**
3	Write a node js script to write the text “You are creating a file” to help.txt file. After that append the text “you are appending data” to same help.txt file. After that read the file and print file contents on console. After finishing read operation , print the line “Thanks for using my program” on console. write ,append,read sequence must be maintain. all read ,write and append operations are asynchronous.	**Unit-2**
4	Write node js script to fetch values from url given below and display output as asked. "https://www.google.com/exam.txt?c1=Hello&c2=FSD2+T1+Test& c3=Welcome+to+LJU#AllTheBest" 1) Data must be written as below in file named “exam.txt”. File name must be fetched from the url given above. Output: Hello! Welcome to LJU FSD2 T1 Test	**Unit-3**

	#AllTheBest 2) Read content from file “exam.txt” and send response to server and display data in “/” page in same format as above but in H1 tag and in red color. 3) If any other page is requested it shows “Page not found” message in plain text.	
5	Write node.js script to print “Welcome to Home Page” with two links containing two pages named as “About Us” and “Contact Us” on home page of server. If user request for About Us page it should display “Welcome to LJ University” in bold font-style with blue color and if user request for Contact Us page it should display “Email:abc@ljinstitutes.edu.in” in italic font-style with red color if any other request is requested it shows “Page not found” message in plaintext.	**Unit-3**
6	write an express.js script to load an HTML file having username and password and submit button. On clicking submit button. It should jump on "check" page using "POST" method. If username is "admin" , then jump on next middleware to print "welcome… admin" , if username is not "admin" , then stay on same middleware to print "warning msg" in red color.	**Unit-4**
7	You have been assigned to develop a user feedback form for a website using Express.js and cookies. Implement the following requirements: Process a form with the following fields: Name, Email , Message , Rating (radio buttons: Bad, Average, Good, Very Good, Excellent) When the user submits the form, store their feedback information (name, email, message, and rating) in a cookie named "feedback" that expires in 10 seconds. Display a confirmation message to the user after successfully submitting the form & Create a link to display the feedback details stored in the "feedback" cookie. When the user click to the link, retrieve the feedback information from the cookie and display it on the page also include a link on the feedback details page to Logout. When the user clicks the link, user redirected to home page. Make app.js file use get method in express js. No need to write html file having form elements.	**Unit-5**
8	Write an expressJS code in which RESTapi is created for json object named Places I love which contains name,country,state,city and rating out of 10(no decimal points) is given.upon passing ratings on the browser it should display the places having that rating.i.e. on localhost:30001/a/10 should display all the places having 10 ratings.	**Unit-5**
9	Write an express js script that allows only image type file to be uploaded using the multer middleware and saves the file to the specific directory called “IMAGES”.	**Unit-6**
10	Write a code to set up nodemailer in Express.js. Sender email id: “lju@gmail.com”. Receiver email ids: “student@gmail.com and faculty@gmail.com”. Mail subject should be “LJ University” Mail body contains “Welcome Student” in h3 tag and in table display data Date 28/06/23, Exam name FSD-2.	**Unit-6**
11	Create react app using function component to perform tasks as asked. First create files as asked below in folder named “ component” 1. Login.js - for the login page content 2. Product.js - for the Product page content 3. Contact.js - for the contact page content 4. About.js - for the about page content Create Home.js file which contains Links for Login,Product,Contact and About page. Also, add functionality of page routing. Finally call Home.js in App.js. Write code for All necessary files.	**Unit-7**
12	Create a React function component named File1 that receives two props, productName and price. The component should display the product name and price. Additionally, add a button labeled "Add to cart" to the component. When the button is clicked onClick event triggered and an alert should be displayed with the message "Your Product is Added to cart!".	**Unit-7**
13	Create react app to craete exam form. Fields are as under. Name -text, Email -email Password -password, Gender-radio, Exam date-datepicker, Exam center-dropdown) Use useState hook to save the state of the form. Also, add validation for the email and password fields. Display submited values.	**Unit-8**
14	Create a React app with three components: parent component P.js and two child components C1.js, C2.js. Use useContext hook to pass two colors from the Parent component P.js to the Child component C2.js. (Yellow color as font color and Blue color as background color) In Child component C1.js import child component C2.js . In C2.js display “Full stack Development” text in h1 heading with above mentioned background color and font color using context.	**Unit-8**
15	Create a collection name Student. Having fields name, age, std, percentage. 1)insert 3 random records in table. 2)find name of all student age>5. 3) update the std for all by 1. 4) arrange all the record of descending order of age 5)show all the name of student who is the oldest student of all records.	**Unit-9**

16	Write a node.js script to define a schema having fields like name,surname,email,password,city. Apply following validations: (1)name field must required . (2)surname must accept values in small letters. (3) Perform Email ID validation on Email field. (4) password must have minimum and maximum length should be 8 & 12 respectively (5)city allowed values are “baroda”,”surat” and “ahmedabad” only	**Unit-10**
17	**Consider a collection student having documents like this:** [{_id:123433,name: "DDD",age:32}, {_id:123434,name: "BBB",age:20}, {_id:123435,name: "AAA BBB",age:10}] **(1)** Retrieve all records having “BBB” as a substring in name by indexing. Apply required indexing. Predict the values of nReturned & docsExamined properties also. **(2)** Create an index and fire a command to retrieve documents having age>15. Stats must return values nReturned=2 & docExamined=2, even though total 3 records are there.	**Unit-10**
18	Create a React form which accepts Name,Rollno and totalmarks,on submitting the form it should be saved in data table inside MongoDB database named student.	**Unit-10**

Major Components/ Equipment	
Sr. No.	**Component/Equipment**
1	Computer
2	Software: code::blocks, Editor, Various packages to run Node.js, express.js and React.js

Course Outcome	
1	Understand the concepts of backend scripting language node.js.
2	Explore the new features of node.js using its framework as express.js
3	Apply React, routing, hooks, and APIs to build dynamic single-page applications.
4	Apply MongoDB and Mongoose to build and integrate database-driven MERN applications.
Suggested Reference Books	
1	The ultimate beginner’s guide to learn Node.js strp by step, John Bach & Alexander Aronowitz, 3rd edition, mEm Inc
2	Pro express.js, Azat Mardan, Apress publication
3	Full stack react, The complete guide to react.js, Anthony Accomazzo et.al.
4	MongoDB, The definitive guide, Kristina Chodorow, Michael Dirolf, O’Reilly publication
List of Open Source Software/Learning website	
1	www.geeksforgeeks.org
2	www.tutorialspoint.com
3	www.programiz.com
4	www.codingninjas.com

Practical Project/Hands On Project		
Sr. No.	**List of Practical Projects**	**Linked with Unit**
1	Create react app using function component to perform tasks as asked. First create files as asked below in folder named “ component” • Login.js - for the login page content • Product.js - for the Product page content like image, product name, price. • Contact.js - for the contact page content (email id and contact number) • About.js - for the about page content (Heading and paragraph) Create Home.js file which contains Links for Product,Contact and About page. Also, add functionality of page routing. Finally call Home.js in App.js.	**7**
2	**Write express js script to meet following requirements:** (1) Create an index.html file & open it on localhost. (2) After clicking submit button, it should jump to "savesession" page. Store username in session. (3) After saving session, redirect to "fetchsession" page & read session value. Put a logout link button here. (4) Jump on "deletesession" page on clicking "logout" link. (5) Destroy the session on this page & redirect to index.html page.	**5**
3	Write a code snippet to configure the multer middleware to store uploaded files in a specific directory called "uploads"	**6**
4	Connect mongodb with nodejs/express using mongoose. Write queries to perform following tasks on employee collection having fields name, age, position, salary using mongoose. 1) Insert 3 documents with random data of having fields name, age, position, salary. 2) Count all employees where age is not equal to 35 and position is “Full Stack Developer”. 3) Update position of all employees to “Software Developer” whose name is “XYZ” and age is equal to 31. If not such document available than insert the document with updated values. 4) Display position and name of the employee having lowest salary.	**10**

5	Create a React app with three components: parent component P.js and two child components C1.js, C2.js. Use useContext hook to pass two colors from the Parent component P.js to the Child component C2.js. (Yellow color as font color and Blue color as background color) In Child component C1.js import child component C2.js . In C2.js display “Full stack Development” text in h1 heading with above mentioned background color and font color using context.	**8**
6	Write a program to develop a ToDO list using reactJS. -Add text field and add button. -Also diplay added task with delete button.	**8**